

1. Determine the mod-4 & mod-3 reductions of length 8 signal vector

$$x = [1, 2, -2, 3, 4, -2, -1, 1]^T$$

Soln: $N=4: [5 \ 0 \ -3 \ 4]^T$; $N=3: [3 \ 7 \ -4]^T$

2. Let $x = [1, 2, 2, 1, 2, 1, 1, 2]$. Compute 4-point DFT of x using the definition in matrix form.

→ Recompute it by first reducing x modulo 4 & then computing 4-DFT of the result.

→ Finally, compute 4-point IDFT of the result & verify that you recover the mod-4 wrapped version of x .

Soln: $[12 \ 0 \ 0 \ 0]^T$; IDFT: $[3 \ 3 \ 3 \ 3]$

3. Compute 8-point FFT of length 8 signal x which is first 8 samples of $x(n) = 4\cos(\frac{\pi n}{2}) + \cos(\pi n)$.

Discuss whether the 8 computed FFT values accurately represent the expected spectrum of $x(n)$.

Soln: $[0 \ 0 \ 16 \ 0 \ 8 \ 0 \ 16 \ 0]^T$

4. Determine DTFT of $y_1(n) = \begin{cases} 1 & -N \leq n \leq N \\ 0 & \text{otherwise} \end{cases}$

Using $y_1(n)$ find the DTFT of $y_2(n) = \begin{cases} 1 - \frac{|n|}{N} & -N \leq n \leq N \\ 0 & \text{otherwise} \end{cases}$

5. Let $g(n)$ & $h(n)$ be two finite sequences of length 7 each. If $y_L(n)$ & $y_C(n)$ denote the linear & 7-point circular convolutions of $g(n)$ & $h(n)$, respectively, express $y_C(n)$ in terms of $y_L(n)$.

6. Let $X(k)$ be a 14-point DFT of a length 14 real seq. $x(n)$. The first 8 samples of $X(k)$ are given by

$$X(0) = 12, X(1) = -1 + j3, X(2) = 3 + 4j, X(3) = 1 - j5, \\ X(4) = -2 + 2j; X(5) = 6 + j3, X(6) = -2 - j3, X(7) = 10.$$

Find remaining samples of $X(k)$. Evaluate following functions of $x(n)$ without computing IDFT of $X(k)$

(a) $x(0)$, (b) $x(7)$, (c) $\sum_{n=0}^{13} x(n)$, (d) $\sum_{n=0}^{13} |x(n)|^2$
 (e) $\sum_{n=0}^{13} e^{j(4\pi n/7)} x(n)$.

Soln: (a) $32/14$ (b) $-12/14$ (c) 12 (d) $498/14$ (e) $-2 - j2$

7. A length-8 sequence is given by $\{x(n)\}$; $0 \leq n \leq 7$ as $x(n) = \{-4, 5, 2, -3, 0, -2, 3, 4\}$, with an 8-point

DFT given by $X(k)$. Without computing IDFT, determine sequence $y(n)$ whose 8-point DFT is given by

$$Y(k) = W_4^{3k} X(k).$$

Soln: $y(n) = \{2, -3, 0, -2, 3, 4, -4, 5\}$

8. Let $X(z) = \sum x(n)z^{-n}$; where $x(n) = (0.4)^n u(n)$.

8. Let $X(z) = Z\{x(n)\}$; where $x(n) = (0.4)^n u(n)$.

(a) Find $g(n) = Z^{-1}\{X(z^2)\}$ without computing $X(z)$.

(b) Find $y(n) = Z^{-1}\{(1 + Z^{-1})X(z^2)\}$ without computing $X(z)$.

9. Let $X(z) = Z[x(n)]$; $x(n)$ is length 12 sequence, starting at $n=0$ and $X_0(k) = X(z)|_{z=e^{j2\pi k/9}}$; $k=0, \dots, 8$.

Determine the 9-pt IDFT $x_0(n)$ of $X_0(k)$ without computing $X_0(k)$

10. Let $X(e^{j\omega}) = \text{F.T.} \left\{ \underset{\substack{\uparrow \\ n=0}}{1}, 1, \dots, \underset{\substack{\uparrow \\ n=7}}{1} \right\}$

$X_1(k)$ is obtained by sampling $X(e^{j\omega})$ at uniform intervals of $\pi/5$ starting from $\omega=0$; i.e., $\omega = \frac{2\pi k}{10}$.

Determine IDFT of $[X_1(k)] = x_1(n)$ without computing $X(e^{j\omega})$ and $X_1(k)$.

Can you recover $x(n)$ from $x_1(n)$.

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